

Thyroid Physiology

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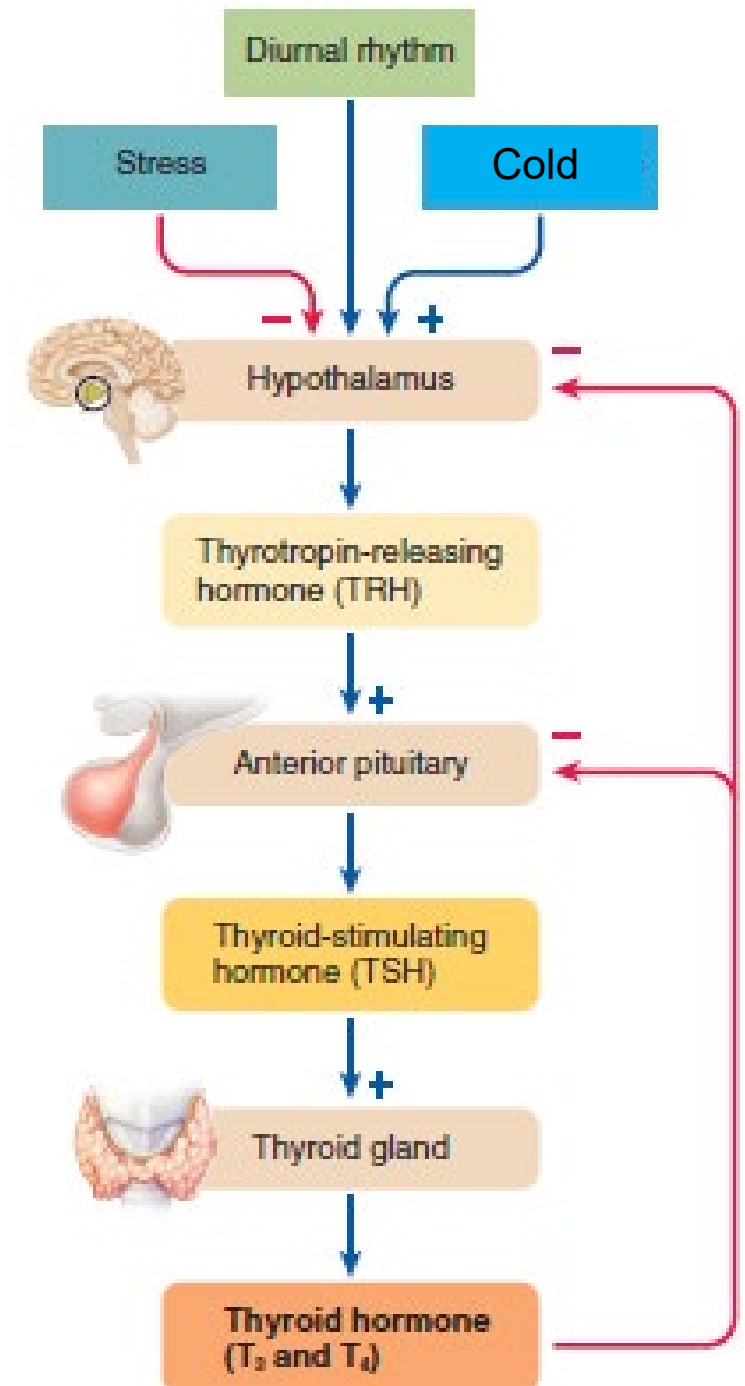
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Learning Outcomes

1. Draw and describe the hypothalamic-pituitary-thyroid axis and its control.
2. Describe regulation of thyroid hormone secretion regulated by the hypothalamic-pituitary-thyroid gland axis.
3. List the most important metabolic and other functions of thyroid hormones.
4. Describe the general and tissue-specific physiological effects of thyroid hormones.
5. Compare the signs of thyroid hormone deficiency in adults and children.
6. Explain the most important mechanism for goiter in thyroid disorders.
7. List the most common thyroid disorders, their causes, and clinicopathologic manifestations.

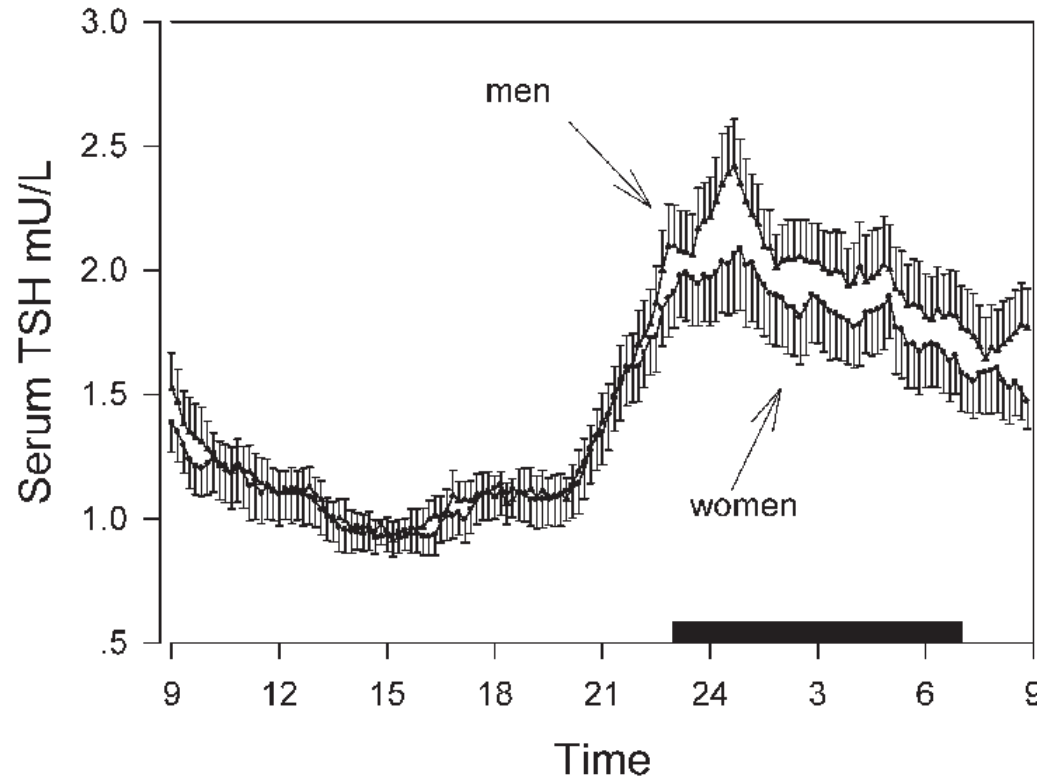
Thyroid hormone is regulated by the hypothalamic-pituitary-thyroid axis

- Hypothalamic neurons release **Thyrotropin Releasing Hormone, TRH**.
 - *TRH is also called thyroid hormone releasing hormone.*
- TRH stimulates the anterior pituitary thyrotrope cells to release **Thyroid-Stimulating Hormone, TSH**.
 - *TSH is also called thyrotropin.*
- TSH stimulates thyroid follicle cells to release thyroid hormones, **T₃** & **T₄** (triiodothyronine & tetraiodothyronine or thyroxine).
- T₃ & T₄ feedback in a classical negative feedback fashion to inhibit TRH & TSH release.



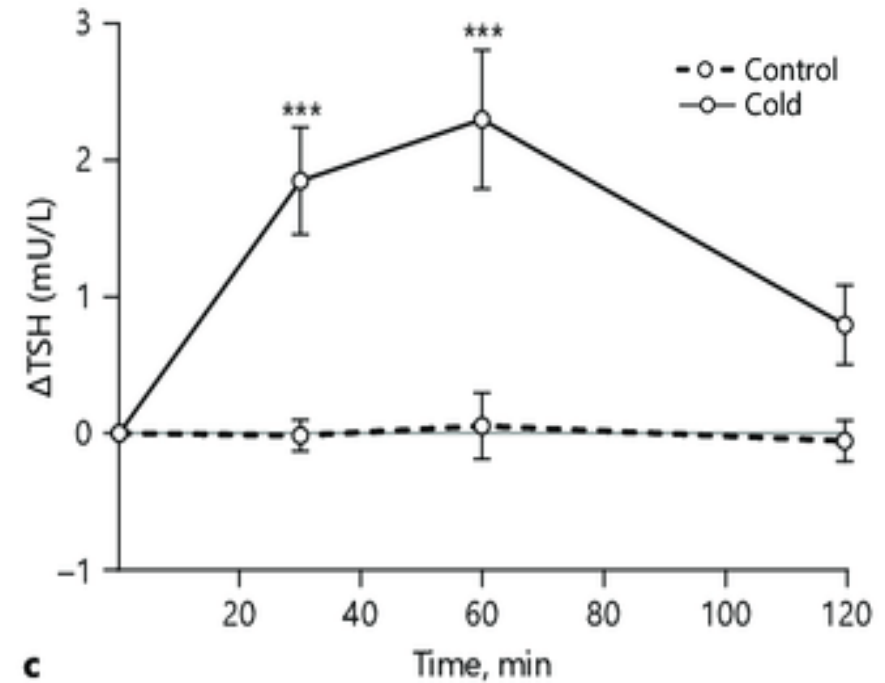
Regulation of thyroid hormone secretion

Circadian rhythm



24 hr serum TSH concentration time series in 24 healthy men and 22 healthy women. Blood samples were taken every 10 min for 24 h. Blood sampling started at 09.00. Lights were off between 23.00 and 07.30 next morning. Data are shown as the group mean and SEM.

Effect of cold exposure on TSH



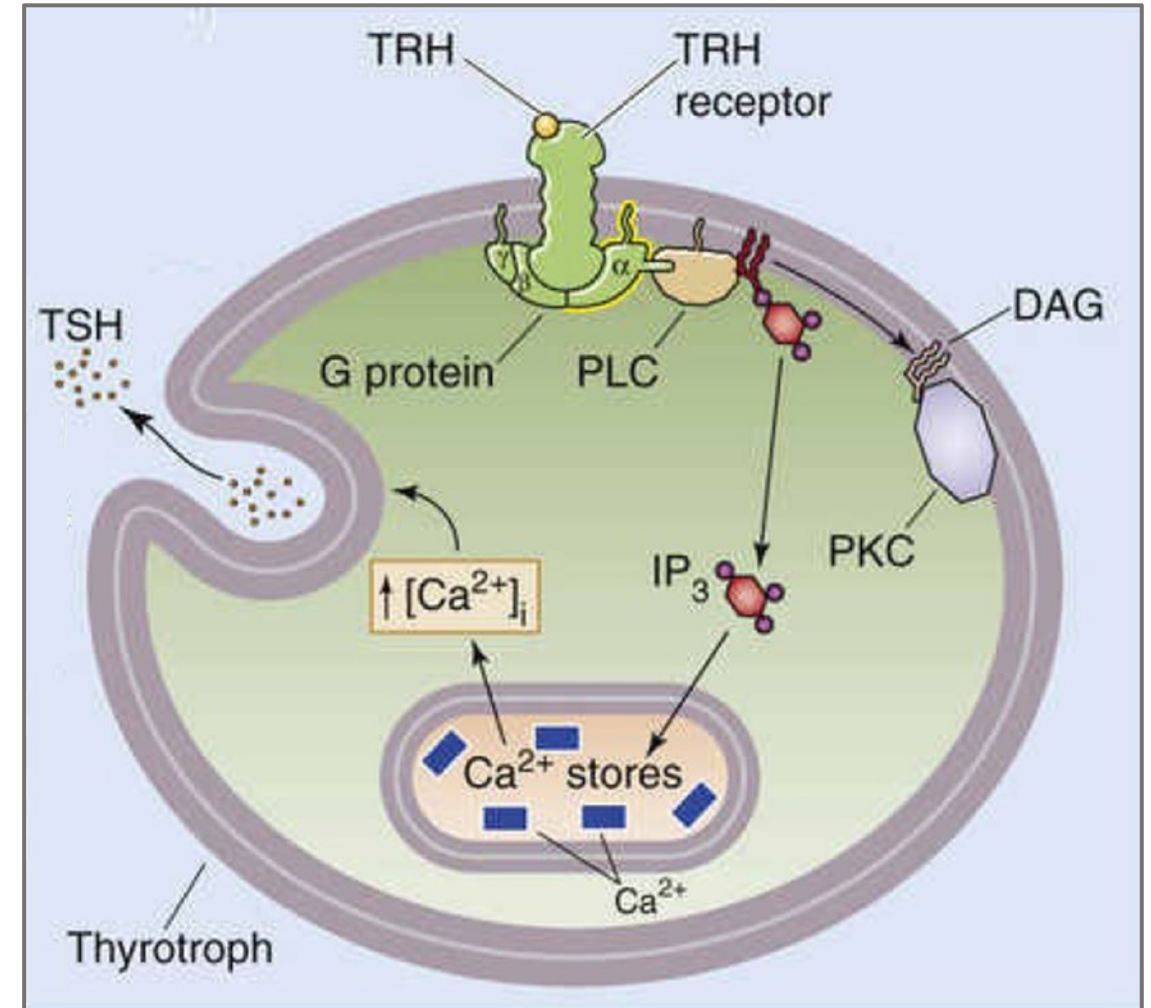
Effects of cold exposure on plasma TSH concentrations at 0, 30, 60, and 120 min after starting the cold exposure. Data are shown as the absolute increase (delta) compared to the basal samples before treatments.

TRH – Thyroid Hormone Releasing Hormone (*aka Thyrotropin Releasing Hormone*)

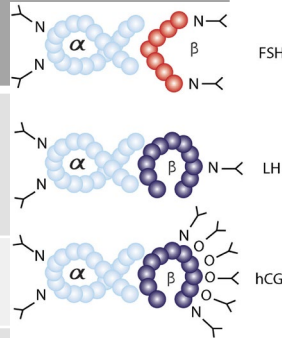
Nature of hormone, chemistry	Peptide
Secreted from	Hypothalamic parvocellular neurons extending from the paraventricular nucleus (PVN) to the median eminence; released into the hypophyseal portal system.
Stimulus for release	Tonically secreted; inhibited by T_3 , T_4 . Stress, anxiety, sympathetic nervous stimulation: $\downarrow$ TRH. Cold: $\uparrow$ TRH.
Half-life	6 mins
Target tissue	Anterior pituitary thyrotropes.
Receptor	Thyrotropin-releasing hormone receptor (TRHR) (Gq GPCR).
Function/physiological effect	Stimulates synthesis & release of thyroid-stimulating hormone (TSH)..
Feedback regulation	T_3 & T_4 exert negative feedback inhibiting TRH release.
Hyposecretion	Insufficient stimulation of the anterior pituitary thyrotropes, $\downarrow$ TSH release.
Hypersecretion	Hyperstimulation of the anterior pituitary thyrotropes, $\uparrow$ TSH release.

TRH stimulates TSH synthesis & release from pituitary thyrotropes

- TRH binds to its Gq GPCR on the thyrotroph membrane.
- This activates phospholipase C (PLC) resulting in hydrolysis of phosphatidylinositol bisphosphate (PIP_2) & generation of inositol triphosphate (IP_3) and diacylglycerol (DAG).
- Stimulation of this pathway leads to protein phosphorylation and a rise in $[\text{Ca}^{2+}]_i$.
- These pathways stimulate the thyrotrophs to synthesize and release TSH.



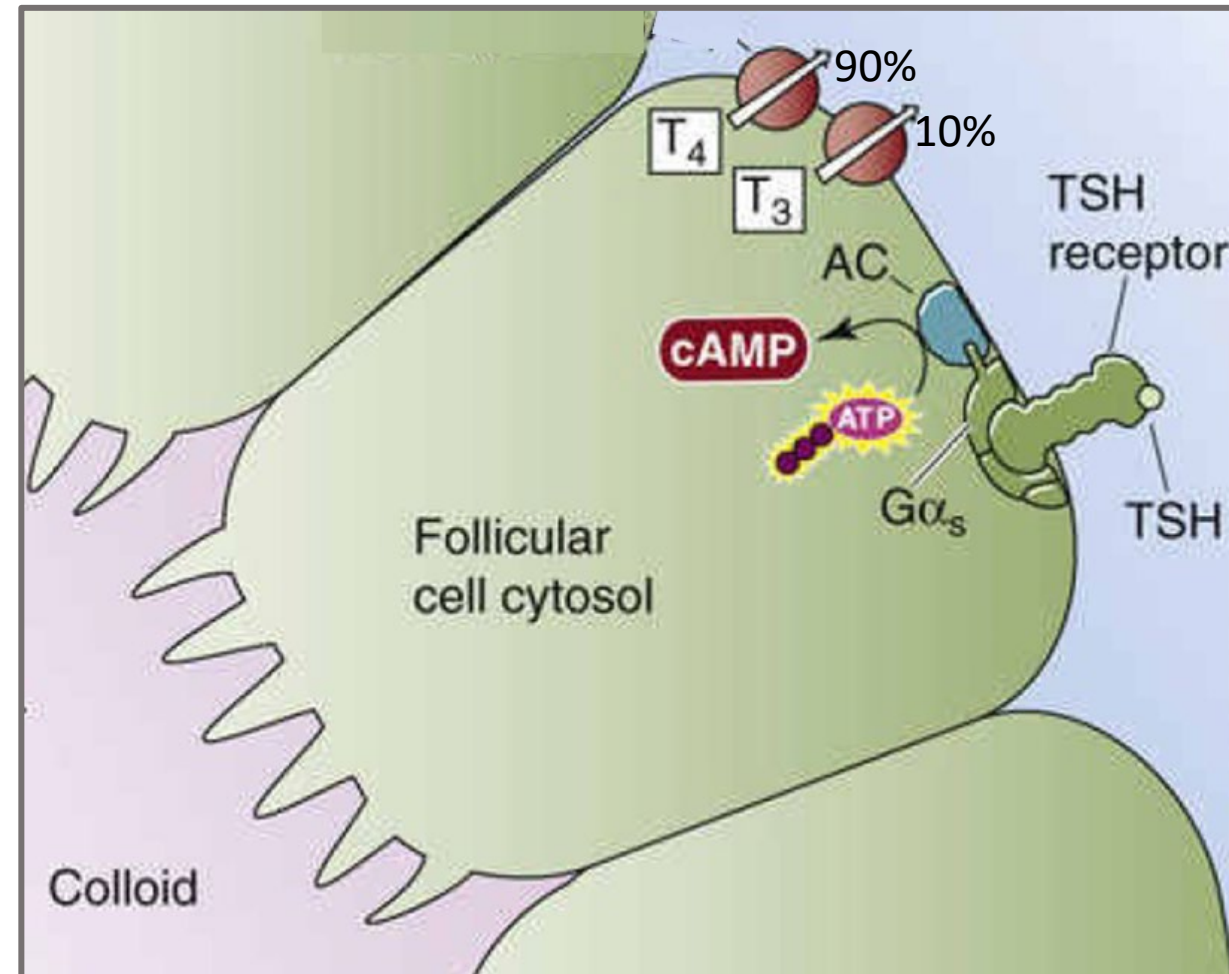
TSH – Thyroid Stimulating Hormone (*thyrotropin*)



Nature of hormone, chemistry	Glycoprotein, consisting of an alpha & beta subunit. Structurally similar to LH, FSH & hCG; shared alpha subunit, distinct beta subunits.
Secreted from	Anterior pituitary thyrotroph cells.
Stimulus for release	Hypothalamic TRH
Plasma binding, Half-life	Unbound, 60 mins
Target tissue	Thyroid follicular cells
Receptor	Thyrotropin-stimulating hormone receptor (TSHR) (Gs GPCR)
Function/physiological effect	Increased thyroid hormone synthesis & secretion; increased thyroid gland growth and blood flow; Prolonged TSH exposure results in follicular hyperplasia.
Feedback regulation	T_3 & T_4 exert negative feedback inhibiting TSH (and TRH) release.
Hyposecretion	Insufficient stimulation of the thyroid follicles: $\downarrow T_3$ & T_4 release, $\downarrow$ thyroid gland size.
Hypersecretion	Hyperstimulation of the thyroid follicles: $\uparrow T_3$ & T_4 release, $\uparrow$ thyroid gland size.

TSH stimulates T_3 & T_4 synthesis & release from thyroid follicles

- TSH binds to its Gs GPCR on the basolateral membrane of thyroid follicular cells.
- This activates adenylate cyclase, raising cAMP levels, stimulating:
 - Iodide uptake
 - Synthesis & secretion of T_3 & T_4
 - Hyperplasia of the thyroid gland
- ~90% of hormone released from thyroid is T_4
- > 99% of T_4 & T_3 in circulation are bound to plasma proteins (*thyroid-binding globulin (TBG), albumin & transthyretin*).
- Unbound, free hormone accounts for biological activity.
 - T_3 is the active form of thyroid hormone.

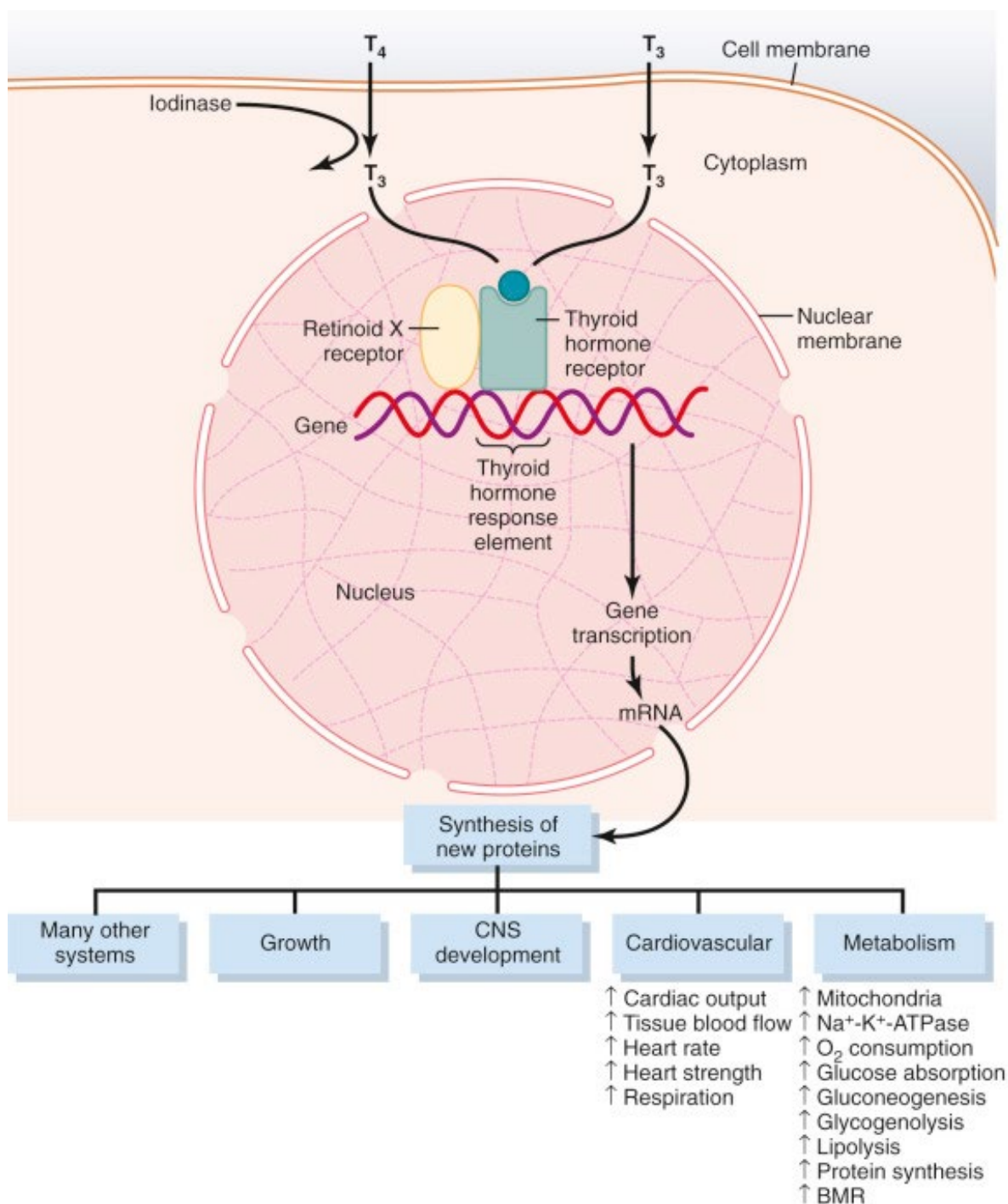


Thyroid Hormone: T₃ (Triiodothyronine) & T₄ (Tetraiodothyronine)

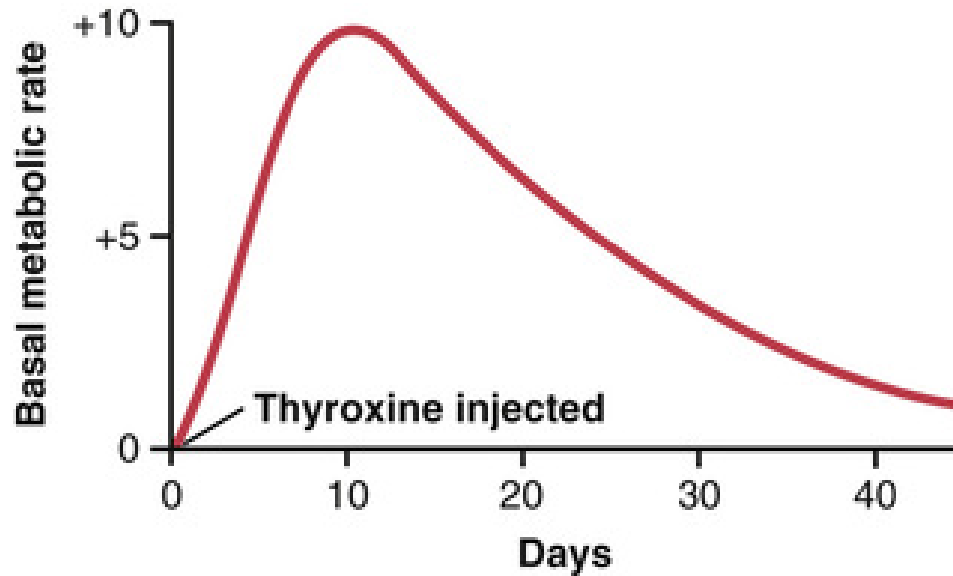
Nature of hormone, chemistry	Iodinated amines. Synthesized from tyrosine and iodine. T ₃ : Triiodothyronine; T ₄ : Tetraiodothyronine/thyroxine.
Secreted from	Thyroid follicular cells (90% T ₄ , 10% T ₃).
Stimulus for release	Pituitary thyroid stimulating hormone (TSH) which is stimulated by hypothalamic thyroid releasing hormone (TRH).
Feedback regulation	T ₃ & T ₄ exert negative feedback at the levels of the anterior pituitary & hypothalamus.
Transport, half-life (t _{1/2})	Bound tightly to thyroxine binding globulin (TBG), transthyretin & albumin. Prolongs half life: ~7 days for T ₄ , ~24 hr for T ₃ .
Receptor signaling mechanism	INTRACELLULAR (NUCLEAR) RECEPTOR**
Target tissue(s)	Most cells of the body.
Function/physiological effect	T ₃ binds to receptor with 10x higher affinity than T ₄ ; ↑ BMR, growth, CNS development, ↑ heat production, ↑ heart rate & cardiac output.
Hyposecretion	Fatigue, cold intolerance, weight gain, constipation, weakness, depression.
Hypersecretion	Difficulty sleeping, heat intolerance, weight loss, anxiety, irritability.

Thyroid hormone acts through nuclear receptors in target tissues

- Free extracellular T_4 & T_3 are transported into target cells.
- Once inside the cell, T_4 is converted to T_3 by a cytoplasmic deiodinase.
- In the nucleus, thyroid hormone receptors (TR) are already bound to nuclear DNA at thyroid response elements (TRE) in the promoter region of certain genes.
- T_3 binding activates TR, thereby regulating transcription of these genes & protein synthesis.



Thyroid hormone has slow onset and long duration of action



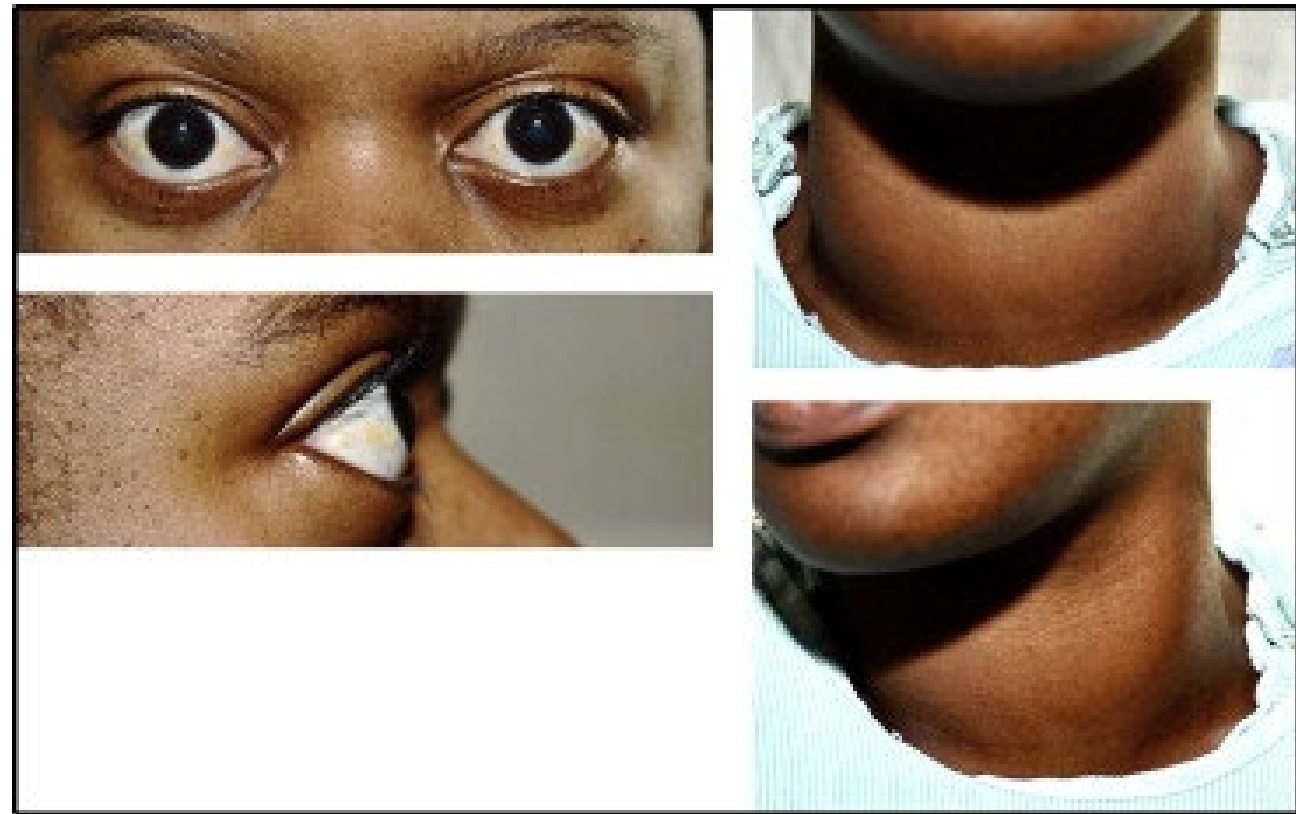
Approximate prolonged effect on the basal metabolic rate caused by administering a single large dose of thyroxine.

- Thyroid hormone stimulates basal metabolic rate (BMR).
- Following injection of large dose of thyroxine (T_4) to a human:
 - 2-3 days before change is observed (*latent period for genomic effect*).
 - 10-12 days before peak is reached.
 - 2 months to return to baseline.
- Prolonged duration of action of THs likely due to binding to carrier proteins & genomic effect.

Patient 1: A 28-year-old female visits her physician with complaints of 20 lb weight loss in 3 months despite an insatiable appetite. She also complains of nervousness, sleeplessness, heart palpitations, and irregular menstrual periods. She notes that she is “always hot”.

On physical examination, the patient is restless and had a noticeable tremor in her hands. She has a wide-eyed stare, and her lower neck appears full. BMI is 16. Heart rate: 110/min. Blood pressure: 120/80 mm/Hg. Resp rate: 24 breaths/min.

Labs reveal **high** TH levels.



Patient 2: A 36-year-old woman presents to the clinic with complaints of several months of generalized weakness, cold intolerance and low mood. She also complains of constipation and heavy menstrual bleeding every month. She has been following a strict plant-based diet for 4 years and avoids all processed/ packaged foods. She has gained weight despite “no interest in food”.

Physical examination reveals hair loss, puffiness in the face and dry skin . Her lower neck appeared full. BMI: 27 kg/m². Heart rate: 50/min. BP: 100/70 mm Hg. RR: 10 breaths/min

Labs reveal **low** TH levels.



Overview of thyroid hormone effects

System	Effects
Cardiovascular	Increases heart rate Increases the force of cardiac contractions Increases cardiac output as a result of the previous two effects Promotes peripheral vasodilation
Central nervous	Essential for normal brain development, such as cerebellar growth and nerve myelination Necessary for normal intellectual development in infants Necessary for emotional stability in adults
Gastrointestinal	Increases appetite Increases secretion of "digestive juices" Increases gastric motility
Hematopoietic	Influences erythropoiesis
Metabolic	Profoundly affects oxidative metabolism Increases oxygen consumption in all tissues except the brain, gonads, and spleen Promotes heat production Influences synthesis and degradation of carbohydrate, fat, and protein
Respiratory	Influences lung development Necessary for surfactant production Increases rate and depth of respirations
Skeletal	Indirectly promotes growth formation by actions on the pituitary gland Acts synergistically with growth hormone and other growth factors that promote bone formation Directly affects skeletal maturation Necessary for progression of tooth development and eruption
Skin	Necessary for growth and maturation of the epidermis and hair follicles

Permissive effects of hormones

- **Permissiveness:** one hormone cannot fully exert its effects unless a second hormone is present.
- Thyroid hormone is an example of a permissive hormone required for full effect of reproductive hormones in reproductive development.

Thyroid hormone alone	→	No development of reproductive system
Reproductive hormones alone	→	Delayed development of reproductive system
Reproductive hormones + adequate thyroid hormones	→	Normal development of reproductive system

Note: The molecular mechanisms responsible for permissiveness are not well understood in most instances.

Thyroid hormones are essential for development of the central nervous system & for normal growth

- The human brain undergoes its most active phase of growth during the last 6 months of fetal life & first 6 months of postnatal life.
- THs are transferred to the fetus during the first trimester from maternal blood, and fetal thyroid becomes active thereafter.
- TH deficiency during these pre- & postnatal periods results in inadequate *brain development*.
- TH promotes *normal growth* by stimulation of GH gene expression : TH deficiency results in GH deficiency.
- TH also promotes *growth & calcification of bones*: TH deficiency in early life leads to abnormal bone development & dwarfism.
- Without TH therapy in deficient children, irreversible impairment of intellectual and physical development can occur – *cretinism*.

Newborn screening programs were developed to detect and treat this condition as early as possible, by measuring either thyroxine (T4) and/or thyrotropin (thyroid-stimulating hormone [TSH]) in heel stick blood specimens.

- Two individuals from the same village in an area of **endemic cretinism : dietary iodine deficiency** leading to **hypothyroidism**
- The girl on the left is a healthy 6-year-old.
- The girl on the right is 17-years-old with **congenital hypothyroidism**:
 - Short stature
 - Obesity
 - Malformed legs
 - Myxedema (puffy face, periorbital edema, dry skin & hair)
 - Enlarged tongue
 - Severe intellectual disability
 - Markedly delayed puberty
 - Muscle weakness
 - Serum T_4 : 0.1 $\mu\text{g/dL}$ (5-12 $\mu\text{g/dL}$)
 - Serum T_3 : 10 ng/dL (115-190 ng/dL)
 - Thyrotropin/TSH: 288 $\mu\text{U/mL}$ (0.5 – 5.0 $\mu\text{U/mL}$)



Effect of thyroid hormone on carbohydrate, fat & protein metabolism

Carbohydrate

- TH stimulates all aspects of carbohydrate metabolism: absorption from GI tract, glucose uptake into cells, glycolysis & gluconeogenesis. Increased blood glucose indirectly increases insulin release.

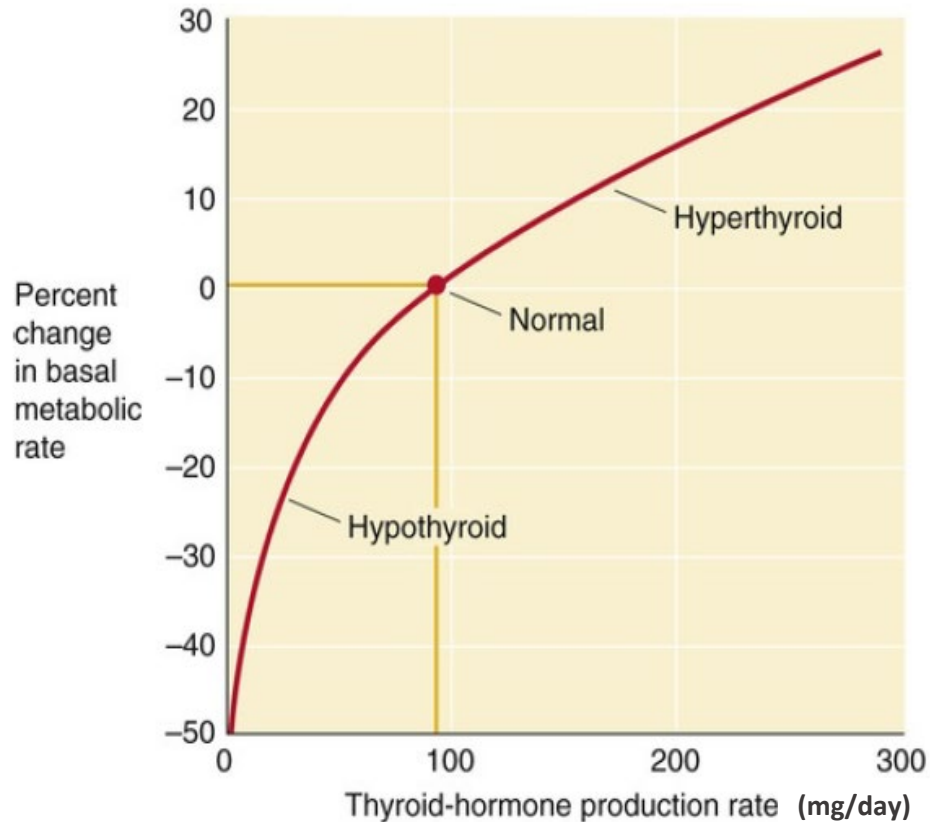
Fat

- TH enhances virtually all aspects of fat metabolism.
- TH stimulates fat mobilization, leading to ↑ serum FA levels & decreased fat stores.
- TH enhances oxidation of FAs in many tissues.
- Plasma concentrations of cholesterol & triglycerides are inversely correlated with TH levels - one diagnostic indication of hypothyroidism is increased blood cholesterol concentration.

Protein

- Protein turnover (release of muscle amino acids, protein degradation & to a lesser extent, protein synthesis) is enhanced by TH. Amino acids used to for hepatic gluconeogenesis. With excess TH, protein degradation outweighs synthesis → loss of muscle protein, weakness.

Thyroid hormone increases basal metabolic rate (BMR)



Effect of thyroid hormone on BMR.

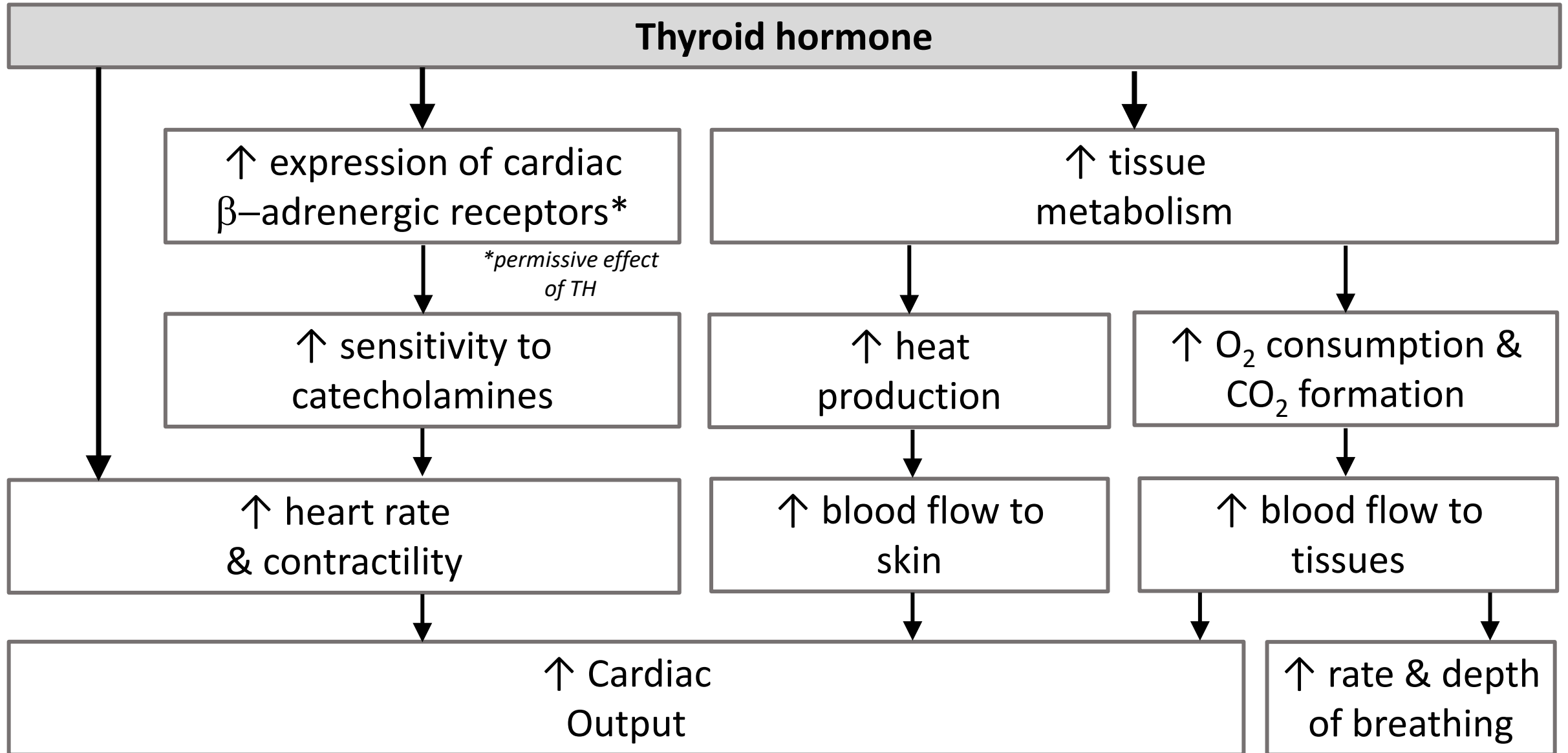
Approximate relation of the daily rate of thyroid hormone (T_4 and T_3) secretion to percent change in basal metabolic rate, compared with normal.

- BMR is the primary source of energy expenditure in humans, primarily accounted for (70%) by those processes required to keep our bodies functioning.
- TH increases BMR by increasing energy expenditure, O_2 consumption & heat production under resting conditions:

1. $\uparrow$ rate of carbohydrate, fat & protein metabolism $\rightarrow$ $\uparrow$ energy consumption.
2. $\uparrow$ expression & activity the Na^+/K^+ pump $\rightarrow$ $\uparrow$ ATP consumption.
3. $\uparrow$ expression of UCP1 (thermogenin) in mitochondria of brown adipose tissue $\rightarrow$ $\uparrow$ O_2 consumption & heat production without ATP generation.
4. $\uparrow$ expression of β -adrenergic receptors* in:
 - Heart ($\uparrow$ contractility), Skeletal muscle ($\uparrow$ carbohydrate metabolism, Na^+/K^+ ATPase activity) & Brown adipose tissue (lipolysis for fueling thermogenesis) $\rightarrow$ $\uparrow$ ATP consumption.

**permissive effect of TH*

Effects of thyroid hormone on cardiovascular & respiratory systems



Excess TH results in increased heart rate, force of contraction, respiratory rate, as well as heat intolerance

Effect of thyroid hormone on sexual function

- Normal reproductive function in males & females requires normal thyroid hormone levels - another permissive effect of TH.
- In females, abnormal TH can result in menstrual cycle dysfunction (lack of menstrual cycles, heavy menstrual bleeding).
- In males, abnormal TH can result in low libido & erectile dysfunction.
- Precise mechanisms uncertain:
 - TH stimulates hepatic production of Sex Hormone Binding Globulin (SHBG)
 - High TRH levels stimulate PRL; hyperprolactinemia → ↓ GnRH → ↓ LH & FSH → ↓ stimulation of the gonads → hypogonadism & infertility.
- Estrogen stimulates TBG
 - Pregnancy & estrogen-containing oral contraceptive pill, ↑ binding of TH in serum → ↓ Free TH → ↑ TRH/TSH → ↑ TH release → ↑ Total Thyroid hormone (bound + free)

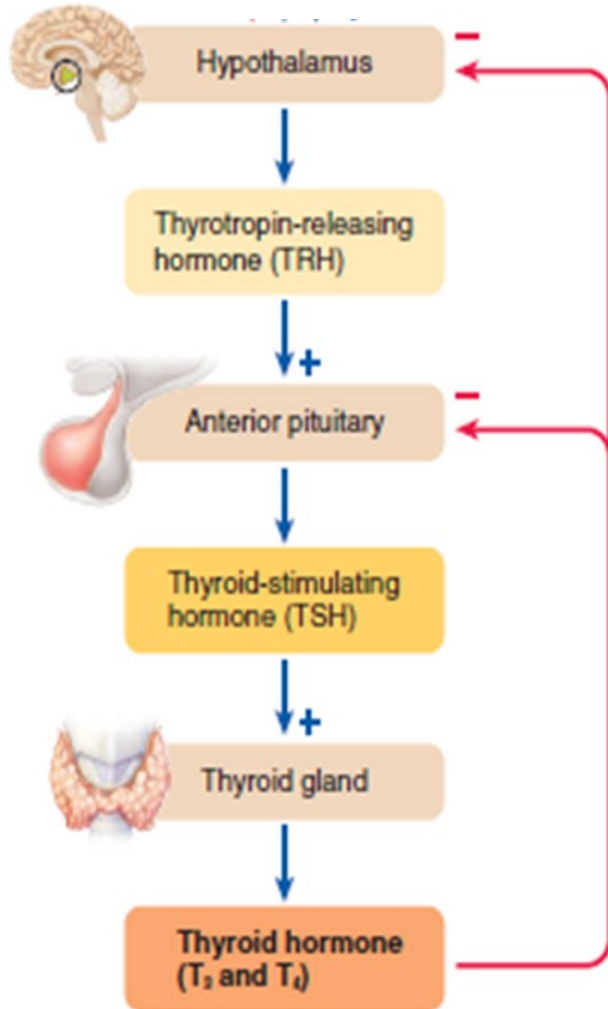
Other effects of TH

- Enhance wakefulness, alertness, responsiveness to various stimuli, auditory sense, awareness of hunger, memory & learning capacity.
- Increase speed & amplitude of peripheral nerve reflexes.
- Increase motility of the GI tract & increase appetite.
- Required for normal cycles of growth & maturation of the epidermis, hair follicles & nails.
- Alter the structure of subcutaneous tissue by inhibiting synthesis & promoting degradation of mucopolysaccharides (glycosaminoglycans) – in hypothyroidism these accumulate resulting in puffy face (myxedema).
- Excess TH results in a fine muscle tremor (important clinical sign).

Thyroid disorders: definitions

- **Euthyroidism** : Normal thyroid hormone and TSH levels.
- **Hypothyroidism**: Under production of thyroid hormone.
 - **Primary** : Failure of the thyroid gland (low TH, high TSH).
 - **Secondary** : Failure of the pituitary gland (low TH, low TSH).
- **Hyperthyroidism**: Over production of thyroid hormone.
 - **Primary**: thyroid gland pathology (high TH, low TSH).
 - **Secondary**: thyroid gland stimulated by excess TSH (high TH, high TSH).
- **Thyrotoxicosis**: clinical manifestation of excess thyroid hormone action at the tissue level due to inappropriately high TH concentrations due to any cause.

Hypothyroidism



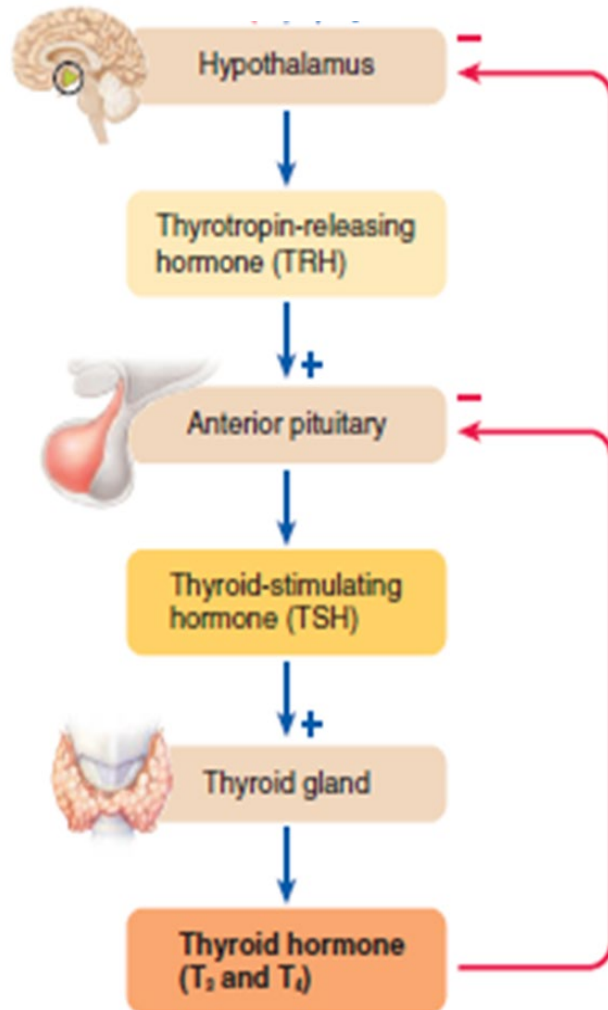
** While TRH levels may be altered in hypothyroidism, it is not typically measured in serum – released into portal circulation*

- **Primary Hypothyroidism** – the thyroid gland is under active & TH production is low ($\downarrow$ TH)
- No negative feedback to hypothalamus/pituitary $\rightarrow \uparrow$ TRH*, $\uparrow$ TSH.
- Causes: chronic iodine deficiency, Hashimoto's thyroiditis (autoimmune destruction of the thyroid gland), thyroid surgery, radiation therapy, over-response to hyperthyroid treatment.

- **Secondary hypothyroidism** – the pituitary gland is under active ($\downarrow$ TSH); thyroid gland is under-stimulated ($\downarrow$ TH).
- No negative feedback to hypothalamus : $\uparrow$ TRH*
- Causes: tumor, trauma, radiation therapy.

- **Tertiary hypothyroidism** – insufficient release of TRH from the hypothalamus, anterior pituitary & thyroid are under-stimulated ($\downarrow$ TRH, $\downarrow$ TSH, $\downarrow$ TH).
- Cause: tumor, trauma, radiation therapy.

Hyperthyroidism



- **Primary Hyperthyroidism** – the thyroid gland is overactive & TH production is high ($\uparrow$ TH)
- Negative feedback to hypothalamus & pituitary $\rightarrow \downarrow$ TRH*, $\downarrow$ TSH
- Causes: Graves' disease, autoimmune disorder in which antibodies specific for TSH receptor on thyroid gland are produced. These thyroid-stimulating immunoglobulins (TSIs) stimulate TH release; Toxic adenoma & toxic multinodular goiter – hyperplasia of thyroid follicular cells – TH release independent of TSH.

- **Secondary hyperthyroidism** – Excess TSH (from pituitary or tumor) ($\uparrow$ TSH); thyroid gland is overstimulated ($\uparrow$ TH).
- Negative feedback to hypothalamus $\rightarrow \downarrow$ TRH*
- Causes: Anterior pituitary tumor, TSH producing tumor.

- **Tertiary hyperthyroidism** – $\uparrow$ TRH from the hypothalamus, anterior pituitary & thyroid gland are over-stimulated ($\uparrow$ TRH*, $\uparrow$ TSH, $\uparrow$ TH). Most typically caused by a tumor.

** While TRH levels may be altered in hyperthyroidism, it is not typically measured in serum – released into portal circulation*

Laboratory tests

USMLE Laboratory values

REFERENCE RANGE

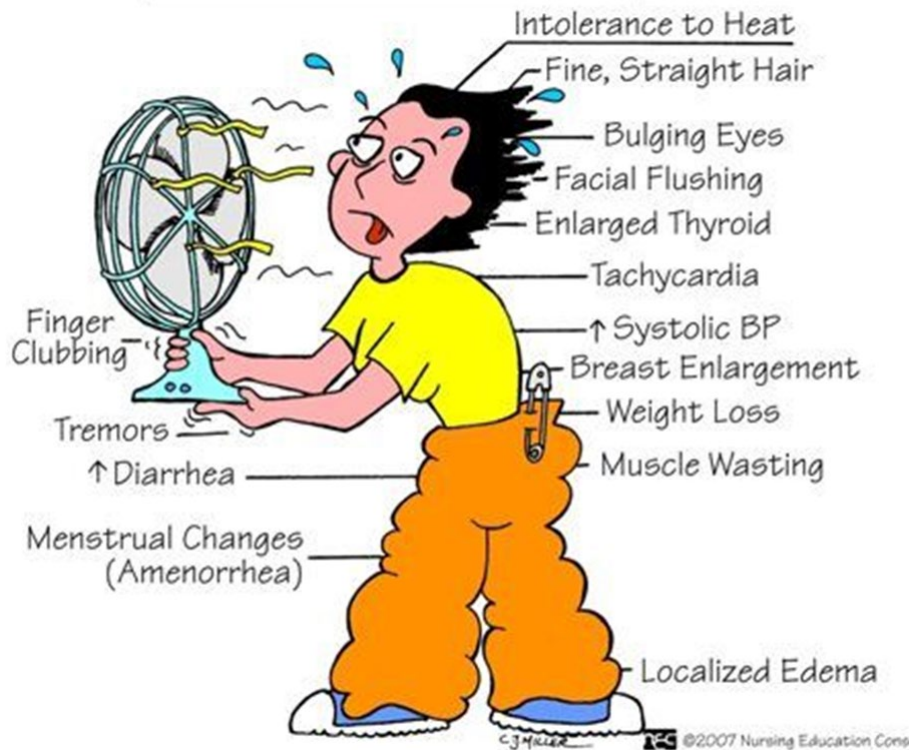
SI REFERENCE INTERVALS

Thyroid-stimulating hormone, serum or plasma.....	0.5-5.0 μ U/mL	0.5-5.0 mU/L
Thyroidal iodine (123 I) uptake.....	8%-30% of administered dose/24 h	0.08-0.30/24 h
Thyroxine (T_4), serum.....	5-12 μ g/dL	64-155 nmol/L
Triiodothyronine (T_3), serum (RIA)	115-190 ng/dL	1.8-2.9 nmol/L

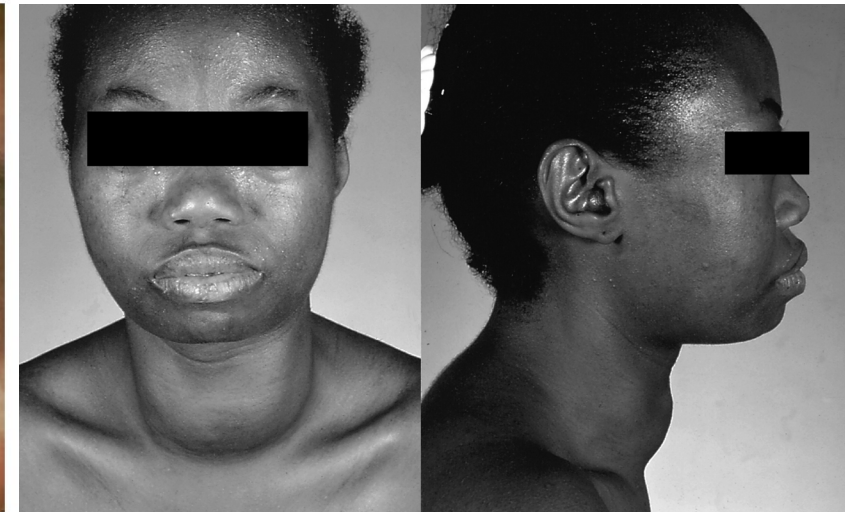
- Most TH in serum is bound to proteins (primarily TBG) & most TH in serum is T_4 .
- Only free TH is biologically active and capable of communicating with the hypothalamus & pituitary, thus regulating TSH secretion.
- Serum TSH assays are both more sensitive and specific than serum free T_4 measurements for outpatients if a single test is utilized, but TSH+ T_4 is more informative.
- Thyroid function tests typically involve measuring TSH and/or free T_4 (FT4, thyroxine) and, to a lesser, extent free T_3 (FT3), depending on the patient presentation.

Graves' Disease

- Most common cause of hyperthyroidism.
- Autoimmune disease in which antibodies (thyroid-stimulating immunoglobulins (TSIs)) form against TSH receptor on thyroid gland.
- TSIs bind to TSH receptor & stimulate TH release : Labs $\uparrow$ TH $\downarrow$ TSH
- Result: Symptoms of TH excess (thyrotoxicosis), exophthalmos, goiter.



Exophthalmos: TSIs stimulate connective tissue in extraocular muscles to synthesize mucopolysaccharides causing increased muscle volume & protrusion of the eyes.



Goiter: Thyroid is usually (not always) diffusely enlarged owing to follicular hyperplasia & hypertrophy resulting from excessive stimulation by TSIs.

Iodine deficiency

- Iodine deficiency is a cause of primary hypothyroidism.
- Evident in developing countries, pregnancy & those on restricted diets.
 - Foods supplemented with iodine are not available (iodized salt), high iodine foods (fish, dairy products) are not regularly consumed, requirements not met.
- Adverse effects on child development.
- In adults, can lead to symptoms of TH deficiency, goiter & myxedema.



Myxedema: accumulation of mucopolysaccharides under the skin. These molecules attract water and cause the puffy appearance.



Goiter: Excessive stimulation of the thyroid gland by TSH causes the gland to enlarge. ($\downarrow$ TH $\rightarrow$ $\uparrow$ TSH)

RDA for iodine

Life Stage	Recommended Amount
Birth to 6 months	110 mcg
Infants 7–12 months	130 mcg
Children 1–8 years	90 mcg
Children 9–13 years	120 mcg
Teens 14–18 years	150 mcg
Adults	150 mcg
Pregnant teens and women	220 mcg
Breastfeeding teens and women	290 mcg

Hashimoto's thyroiditis (chronic autoimmune thyroiditis)

- The most common cause of hypothyroidism in *iodine-sufficient areas*.
- Prevalence increases with age & more common in women than men.
- Characterized by gradual thyroid failure, with or without goiter formation, due to auto-immune mediated destruction of the thyroid gland.
- Most patients have high serum concentrations of antibodies against one or more thyroid antigens (thyroglobulin, thyroid peroxidase).



Signs & symptoms of hypothyroidism

Summary of changes in hormone levels in several thyroid disorders

Diagnosis	Problem	TH	TSH	TRH
Primary hypothyroidism	Underactive thyroid	Decreased	Increased*	Increased*
Secondary hypothyroidism	Underactive pituitary	Decreased	Decreased	Increased*
Tertiary hypothyroidism	Underactive hypothalamus	Decreased	Decreased	Decreased
Primary hyperthyroidism (e.g., Graves' disease)	Overactive thyroid	Increased	Decreased**	Decreased**
Secondary hyperthyroidism	Overactive pituitary	Increased	Increased	Decreased**
Tertiary hyperthyroidism	Overactive hypothalamus	Increased	Increased	Increased

*Due to lack of negative feedback from TH

**Due to negative feedback from TH